ASSIGNMENT – 1

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COURSE: CSA0914-JAVA

Aim

To write a Java program to implement binary search on an array of integers using loops and conditional statements.

Algorithm

1. Start the program.
2. Read the number of elements in the array.
3. Read the array elements in ascending order.
4. Read the element to be searched.
5. Initialize low = 0 and high = n - 1.
6. Calculate the middle index using mid = low + (high - low) / 2.
7. Compare the middle element with the search element:

- If equal, print the element's index.
- If smaller, search the left half.
- If greater, search the right half.

8. Repeat until the element is found or low > high.
9. Display the result.

Program:



```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];

        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
        }

        int key = sc.nextInt();

        int low = 0;
        int high = n - 1;
        int found = -1;

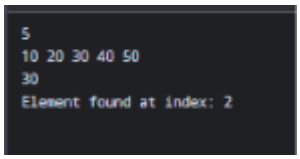
        while (low <= high) {
            int mid = low + (high - low) / 2;

            if (a[mid] == key) {
                found = mid;
                break;
            }
            else if (key < a[mid]) {
                high = mid - 1;
            }
            else {
                low = mid + 1;
            }
        }

        if (found != -1) {
            System.out.println("Element found at index: " + found);
        } else {
            System.out.println("Element not found");
        }

        sc.close();
    }
}
```

Output:



Test Cases

Test case	Array	Search key	Expected output
1	10 20 30 40 50	30	Found at index 2
2	10 20 30 40 50	10	Found at index 0
3	10 20 30 40 50	50	Found at index 4
4	10 20 30 40 50	25	Element not found

Time and Space Complexity

- Best case: O(1)
- Average case: O(log n)
- Worst case: O(log n)
- Space complexity: O(n) for the input array; the search itself uses O(1) extra space.

Result

Thus, the Java program to implement binary search on an array of integers using loops and conditional statements was successfully implemented and executed.